DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Winter Semester Examination – December - 2019

Branch:- Electronics and Telecommunications **Sem:- V**
Subject: - Data Structure & Algorithms Using Java Programming (BTEXPE506C)
Date:- 20/12/2019 **Marks: 60** **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

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(Marks)

Q.1. **Solve Any Two**

a) Explain the types of data structures in detail with suitable diagram (06)

b) Find upper bound, lower bound and tight bound for a function $10n^2 + 4n + 2$ (06)

c) Define Algorithm. Explain characteristics of Algorithm (06)

Q.2. **Solve Any Two**

a) Write an algorithm to convert (06)
  i) Infix expression to postfix expression
  ii) Infix expression to prefix expression

b) What is stack? Explain similarities and differences between stack and queue (06)

c) Covert following infix expression to its equivalent post fix expression using stack algorithm Y = ((P+Q)*S)^(T-U) . (06)

Q.3. **Solve Any Two**

a) Describe various operations associated with linked lists. (06)

b) Write algorithm to delete the node from the linked list (06)

c) Explain algorithm for inserting node in between to singly linked list with suitable example (06)

Q.4. **Solve Any Two**

a) What is complete binary tree? How binary trees are represented in memory? (06)

b) Explain the algorithm for post order traversal algorithm of binary tree (06)

c) Suppose alphabets are inserted in order into empty binary search tree as- (06)

S, T, P, Q, M, N, O, R, K, V, A, B. Find the Final Tree also find inorder, Preorder and Postorder traversal of tree

Q.5. a) Explain Quick Sort Algorithm in detail. (06)

b) Explain the concept of hashing also various hash functions (06)

Q.6. a) Explain breadth first technique or graph traversal (06)

b) Define graphs and how to represent graph by using adjacency matrix (06)

Paper End